

Mathematical Formulation of Net Energy Probabilities and Grid Failure Risk

1 System Definitions and Terminology

Let t denote a discrete time step over interval Δt . We define the net grid balance E_t as a composite continuous random variable:

$$E_t = G_t + B_t - D_t = W_t + S_t + B_t - D_t \quad (1)$$

where:

- $W_t \geq 0$: Wind power output with Probability Density Function (PDF) $P_W(w)$.
- $S_t \geq 0$: Solar power output with PDF $P_S(s)$. During nighttime hours, $P_S(s) = \delta(s)$ (Dirac delta centered at zero, yields $S_t = 0$).
- $G_t = W_t + S_t$: Aggregated renewable energy generation.
- B_t : Active net power supplied by ($B_t > 0$) or absorbed by ($B_t < 0$) the battery reservoir, governed by PDF $P_B(b)$.
- $D_t > 0$: Consumer demand load with PDF $P_D(d)$.
- E_t : Net energy surplus ($E_t > 0$) or deficit ($E_t < 0$).

2 Battery Reservoir and Discharge/Charge Dynamics

Let $SoC_t \in [0, Cap]$ represent the State-of-Charge (stored energy reservoir) at time t . Over a duration Δt , the available discharge power capability B_{dis} and maximum charging power absorption B_{ch} are constrained by power ratings (B_{max}) and physical reservoir limits:

$$B_t = \begin{cases} \min \left(B_{max}, \frac{SoC_{t-1}}{\Delta t} \cdot \eta_d \right) & \text{if } D_t > G_t \quad (\text{Discharging, } B_t > 0) \\ -\min \left(B_{max}, \frac{Cap - SoC_{t-1}}{\Delta t \cdot \eta_c} \right) & \text{if } G_t > D_t \quad (\text{Charging, } B_t < 0) \end{cases} \quad (2)$$

where $\eta_d, \eta_c \in (0, 1]$ represent discharging and charging efficiencies, respectively. The updated reservoir state evolves according to:

$$SoC_t = SoC_{t-1} - \left(\frac{B_t}{\eta_d} \cdot \mathbb{I}_{\{B_t > 0\}} + B_t \cdot \eta_c \cdot \mathbb{I}_{\{B_t < 0\}} \right) \Delta t \quad (3)$$

3 Derivation of Probability Density Function $f(e)$

Assuming independence among random variables W_t , S_t , B_t , and D_t at time t , the joint probability density function is the product of their individual marginal densities:

$$P(W_t = w, S_t = s, B_t = b, D_t = d) = P_W(w)P_S(s)P_B(b)P_D(d) \quad (4)$$

To determine the probability that net energy E_t equals a specific value e , we substitute $d = w + s + b - e$ into the demand probability density $P_D(d)$:

$$P(E_t = e) = f(e) = \int_b \int_s \int_w P_W(w) P_S(s) P_B(b) P_D(w + s + b - e) dw ds db \quad (5)$$

4 Loss-of-Load Failure Probability $P(E_t < 0)$

The grid failure risk (Loss-of-Load Probability, LOLP) corresponds to all negative net energy realization events ($E_t < 0$). Integrating the marginal deficit function $f(e)$ over all negative values yields:

$$P(E_t < 0) = \int_{-\infty}^0 f(e) de = \int_{-\infty}^0 \left(\iiint P_W(w) P_S(s) P_B(b) P_D(w + s + b - e) dw ds db \right) de \quad (6)$$